Multimodal Nowcasting of Sparse Charge-Exchange Spectroscopy on KSTAR: Fast Diagnostics Recover Ion Temperature Beyond Future-Using Interpolation, but Not Toroidal Rotation

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Draft — August 9, 2026

Abstract

Charge-exchange spectroscopy (CES) is the primary measurement of ion temperature (T_i) and toroidal rotation (V_{rot}) in tokamaks, but photon-collection constraints make it slow and intermittently missing: on the KSTAR 10 ms grid studied here, 8.2% of T_i and 23.9% of V_{rot} samples are absent, the two targets go missing independently, and a further 41.1% of V_{rot} rows are instrument holds, leaving 65.0% of the grid without independent V_{rot} information. We ask whether the always-available fast diagnostics—beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI), and Mirnov coils—together with the irregular past CES history can recover information at missing CES timesteps that *temporal interpolation of CES alone cannot*. We train a compact (0.2M-parameter) multimodal network whose history encoder pools a GRU over the irregular CES history with attention *hard-masked to genuinely observed timesteps*—writing interpolation's own inductive bias into the model at zero parameter cost—together with per-diagnostic encoders, explicit irregular-time features, per-target masked losses, and physics-informed per-target input routing, and evaluate it against a deliberately adverse bar: offline interpolation (linear, monotone-cubic PCHIP, local AR) that is allowed to see *past and future* CES neighbors, while the model is strictly causal. Under a pre-registered protocol—file-level train/val/test splits, a test set never touched by model selection, and a shot-clustered paired bootstrap (10,000 resamples)—the model beats PCHIP on T_i with skill +0.18 to +0.28 and a 95% CI excluding zero on *four independent held-out splits*, while V_{rot} clears the same gate on only one of the four—what noise alone would produce—so we report it as a tie. We then stress the result two ways, and the two agree: reweighting the scored points onto the covariate distribution of the *genuinely missing* ones, and re-splitting strictly in time across campaigns, each remove the advantage over offline interpolation while leaving the advantage over every *causal* baseline intact (+0.29 and +0.22). We therefore separate the two claims rather than conflate them, and trace the campaign-shift loss to a measured cause: the fast diagnostics drift 5–11× more between campaigns than the quantity being predicted. Input ablations show this asymmetry is physical rather than architectural: fast diagnostics alone beat persistence for T_i (collisional e–i coupling makes T_e and n_e informative) but are far worse than the mean for V_{rot} (the dominant NBI torque is unobserved and 100 Hz-sampled Mirnov signals alias away kHz mode rotation). The model's advantage concentrates in high-local-variability neighborhoods where smooth interpolation is weakest. We also document a data-quality audit (54% of observed V_{rot} values are forward-filled instrument holds) and an architecture-selection protocol gated on the clean interpolation-relative skill rather than on validation loss, which turned an initially non-significant result into the significant headline. We release the full pipeline, pre-registration, and evaluation artifacts.

1 Introduction

Ion temperature T_i and toroidal rotation V_{rot} at the plasma edge govern key elements of tokamak performance and stability, including pedestal structure, ELM behavior, and access to high-confinement regimes. The workhorse measurement of both quantities is charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of charge-exchange line emission. Because CES must integrate photons to reach adequate signal-to-noise, it is intrinsically slow, and individual time points are frequently lost to exposure and signal-quality problems. In the KSTAR data studied here, even on a uniform 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—*independently* of each other—while a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit (§3.4), so only 35.0% of the grid carries an independent V_{rot} measurement. Empirically in our dataset, the missingness concentrates at physically interesting moments (low signal-to-noise phases, ELMs, transitions), i.e. it is missing-not-at-random in the sense of Little and Rubin [2] (MNAR).

Meanwhile, several fast diagnostics observe the same plasma without gaps: BES measures density-fluctuation structure, ECEI images electron temperature, and Mirnov coils record magnetic fluctuations. None of them measures T_i or V_{rot} directly. The question this paper asks is:

> *At a 10 ms timestep where CES is missing, can the simultaneous fast diagnostics plus the irregular past CES history recover T_i and V_{rot} information that temporal interpolation of CES alone cannot?*

A positive answer would justify a data-driven *virtual sensor*: a model that fills CES gaps online, without the strong inverse-mapping assumptions required by hardware alternatives, enabling gap-free pedestal analysis and, ultimately, real-time use.

A deliberately hard bar. The obvious baseline for gap-filling is interpolation of the CES channel itself. We make the comparison intentionally adverse to the model: the interpolation baselines (linear, monotone-cubic PCHIP [3], and a local AR reference) are *offline*—they may use CES neighbors on *both sides* of the target time, including the future—whereas the model is strictly causal, seeing only the fast diagnostics at the target step and a short window of past CES. If a causal model beats a future-using interpolant, the fast diagnostics must carry CES-relevant information that the temporal structure of CES alone does not contain; and a model that beats future-using interpolation a fortiori beats every causal baseline (persistence, AR).

Contributions.

1. **A leakage-hardened multimodal nowcasting pipeline** for sparse, irregularly observed CES targets: per-target masked multitask losses that recover the ≈28% of labeled rows discarded by both-targets-required filtering; explicit irregular-time features; full masking of the target timestep in the CES-history input; file-level (shot-level) splits; and train-file-only normalization statistics (§3). Its one non-generic component is **observation-masked attention pooling** (§4): each target's attention readout over the history may place mass only on timesteps where *that* target was genuinely measured, which is exactly the inductive bias that makes interpolation strong, added at zero parameter cost.

2. **A map of where the remaining headroom is—and is not** (§6.7, §9). A 24-run window sweep shows that removing previous-CES entirely drops both targets *below* the interpolation bar, that a *single* past observation recovers the model's whole margin, and that the curve is flat from one observation onward. We do not read this as a ceiling but as a routing instruction: the next gain is not in the estimator, and we identify three specific places it *is*, each with the measurement that would deliver it (§9). One is already demonstrated in our own data—the Mirnov channel is uninformative because 100 Hz decimation without anti-aliasing destroyed it *before* the model, not because the plasma lacks the signal (§6.6).

3. **A pre-registered, shot-clustered evaluation protocol** (§5) with a three-way split whose test set is never read by model selection, and significance assessed by a shot-clustered paired bootstrap—adjacent samples within a discharge are strongly correlated, so the discharge, not the sample, is the unit of replication.

4. **A robust positive result for T_i** (§6): the causal model beats future-using PCHIP interpolation with $\text{skill}_{\text{PCHIP}} = +0.18$ to $+0.28$ and shot-clustered 95% CIs excluding zero on four independent held-out splits, and survives a genuine-measurement-only re-evaluation.

5. **A physics-grounded negative result for V_{rot}**, reported as a finding rather than a failure: fast diagnostics carry essentially no rotation information at 10 ms (unobserved NBI torque; Mirnov aliasing), confirmed by input ablations, so V_{rot} remains history-autocorrelation-limited and ties interpolation (significant on 1/4 splits, which is what noise alone would give). We call this the $T_i \leftrightarrow V_{\text{rot}}$ *information asymmetry*.

6. **Diagnosis of where the model earns its edge**: skill is concentrated in high-local-variability neighborhoods, where smooth interpolation is weakest; there, even V_{rot} shows a (fragile) significant edge, and an ablation shows the T_i edge genuinely uses the multimodal signal.

7. **Two stress tests that separate two claims usually conflated** (§6.4, §6.5). Reweighting the scored points onto the covariate distribution of the genuinely missing ones, and re-splitting strictly in time across campaigns, both remove the advantage over *offline* interpolation (1/4 and 0/4) while leaving the advantage over *causal* baselines intact (+0.29, +0.22). An online virtual sensor competes with persistence, not with an interpolant that reads the future; keeping the two apart is what makes the deployment claim defensible. The campaign loss is then traced to a measured cause—BES and ECEI drift $1.22\,\sigma$ and $0.53\,\sigma$ between campaigns against $0.115\,\sigma$ for the CES targets—which indicts train-file-only normalization and names its repair. The MNAR analysis also yields a scope fact we had not stated: at $W{=}4$ only 54% of missing T_i rows (and 4.8% of V_{rot}) are in the model's domain at all.

8. **Deployability, measured** (§8): p99 inference latency of 6.4 ms on CPU against the 10 ms CES cadence—and the counterintuitive finding that a GPU is 8× *slower* at batch 1 for a model this small—plus distribution-free conformal prediction intervals that beat both baselines' intervals on 8/8 seed-target combinations without retraining or perturbing any reported number.

9. **A complexity ladder** (§6.8) answering "this model is too complex to explain" with a measurement rather than a rebuttal: a fully interpretable 1,258-parameter anchor-plus-correction model recovers 31.5% of the T_i margin and 7% of the V_{rot} margin, so we can state exactly what the remaining 0.2M parameters buy.

10. **Two methodological artifacts**: a data-quality audit revealing that 54% of observed V_{rot} values are forward-filled instrument holds—which we show also contaminate *training*—and an architecture-selection protocol whose gate is the clean interpolation-relative skill rather than validation loss.

2 Related work

Profile fitting and gap-filling in fusion diagnostics. Gaussian-process regression [4] is the community-standard tool for fitting and uncertainty-quantifying fusion profile data [5, 6] — applied to KSTAR's own CES T_i profiles with SVM-based outlier rejection [7] — and monotone interpolants such as PCHIP [3, 8] are preferred over natural cubic splines [9] near transients because splines exhibit Gibbs-like ringing at ELM- and sawtooth-like steps [10]. Irregularly sampled series admit exact AR-type null models (IAR/CIAR) [11, 12]. We use this ladder—persistence, linear interpolation [13], PCHIP, local AR—as pre-registered baselines, with PCHIP as the headline comparator, and Murphy's MSE skill score [14] as the metric. Our contribution is orthogonal to this literature: rather than a better CES-only interpolant, we test whether *other diagnostics* add information beyond any CES-only method, by benchmarking a causal multimodal model against future-using interpolation.

Neural diagnostic inference in fusion. Neural diagnostic inference has three main lines, and it is instructive to ask of each: *what happens when the target measurement is absent?* The oldest line accelerates spectral fitting—NN analysis of JET CX spectra [15, 16] and fast T_i estimation from EAST CXRS spectra [17]. These gain orders of magnitude in speed over least-squares fits, but by construction they require measured spectra at the timestep: a missing CES exposure leaves them with no input at all. A second line replaces one inference pipeline with another diagnostic's: Bayesian-model emulation at W7-X [18], profile shapes from scalar parameters on NSTX-U [19], and—closest to our targets—T_i and rotation profiles from an x-ray crystal spectrometer on EAST [20]. Lin et al. predict exactly our two quantities, but from another Doppler spectrometer, i.e. the same physical information channel with the same photon-statistics limits; the mapping is instantaneous and inherits the source diagnostic's availability. A third line builds end-to-end surrogates for slow reconstruction codes: bolometer tomography [21], line-integral diagnostics [22], safety-factor profiles [23], real-time equilibria [24], and—on KSTAR itself—NN Grad–Shafranov solvers [25, 26] and rapid EPED-parameterized profile reconstruction [27] — the latter consuming CES as an *input* and explicitly naming rotation as future work. These address analysis *latency*, not measurement *availability*: they accelerate inversions of data that exist. Two works sit closer to the availability question. Actuator-conditioned profile forecasting on DIII-D [28] is causal and history-driven like our model, but forecasts from *dense* measured profiles for control, rather than recovering a sparse channel. RTCAKENN [29] reconstructs kinetic profiles (including impurity temperature and rotation) in real time and keeps operating when Thomson/CER inputs drop out, via dropout training; missingness there is a robustness condition handled by zeroing inputs, whereas in our setting the target's own sparsity is the problem itself, tracked explicitly through per-target observation flags and irregular-time features. Across all three lines the common assumption is that the information source is available when the model runs; recovering a diagnostic at timesteps where *its own* measurement is missing, from other channels plus its past, falls outside their scope.

Cross-diagnostic reconstruction and temporal densification (closest prior work). A recent family reconstructs one diagnostic from others. GP-based imputation restores faulty magnetic sensors on KSTAR in sub-ms [30]; ML detects and compensates corrupted sensors [31]; autoencoders fuse redundant density diagnostics for resilience [32]; time-series extrinsic regression fills missing electron-temperature data on EAST [33]; and FusionMAE performs masked-channel "virtual backup diagnosis" across 88 channels on HL-3 [34]. Closest to our framing, Diag2Diag synthesizes Thomson-scattering T_e/n_e at fluctuation-diagnostic rates from other simultaneous diagnostics on DIII-D [35], and temporal super-resolution of Thomson profiles from fast radiation diagnostics has been demonstrated on COMPASS [36]; PanoMHD models multimodal signal dynamics with a causal transformer for control [37]. Our work differs from this family on four axes, each verifiable from the cited works' stated scope: (i) *target channel*—every member reconstructs electron quantities or generic dense channels (Thomson T_e/n_e in 35 and 36; ECE-type T_e in 33; arbitrary masked channels in 34), and where CER/CES appears at all it is an *input* [35], never the sparse target; (ii) *causality*—Diag2Diag and the COMPASS study are simultaneous, memory-less mappings, and FusionMAE reconstructs within a window, whereas our model conditions on the target's own irregular past and never reads the future; (iii) *evaluation bar*—none benchmarks against *future-using* interpolation of the target channel, which for gap-filling is the incumbent method to beat, under a pre-registered shot-clustered protocol; and (iv) *information analysis*—reconstructability is assumed rather than tested per target, whereas the $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry is our central finding. In sum, this work continues the programme this active family has established—reconstructing an absent measurement from the diagnostics that are always on—and extends it in three directions: from electron channels to a *sparse ion channel*, from simultaneous memory-less mappings to *causal estimation conditioned on the target's own irregular past*, and from assumed reconstructability to *per-target tests against pre-registered baselines*. To our knowledge the conjunction of these three—causal temporal gap-filling of sparse CXRS ion measurements—has not yet been addressed; we present it as the family's natural next step rather than as a departure from it.

Irregular-time and masked multitask learning. Our design borrows standard ingredients from the irregular time-series literature—observation masks and decay-aware recurrence (GRU-D [38]), attention over continuous-time embeddings (mTAN [39]), and latent-ODE dynamics [40]—together with multitask learning under partial labels [41], late-fusion multimodal architectures [42, 43], pre-LayerNorm transformers [44], and attention pooling [45]. The "nowcasting" framing follows its use in machine-learning weather prediction [46, 47]. The novelty is not any single ingredient but the leakage-hardened combination under a physics-adverse evaluation protocol.

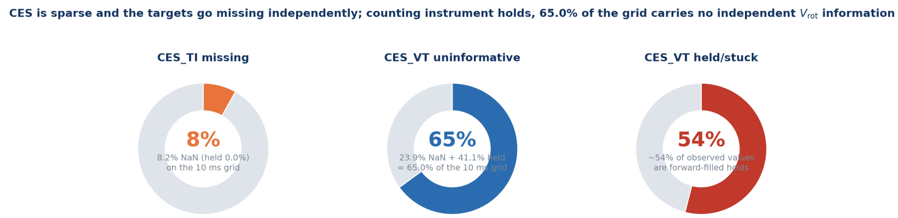

Figure 1: CES missingness structure on the 10 ms grid. T_i and V_{rot} go missing independently (≈8% / ≈24%), motivating per-target masking rather than row filtering.

3 Data and problem setup

3.1 Dataset

We use 641 KSTAR discharge (shot) files, prioritized from campaigns #24000–#33000 for hardware consistency, targeting H-mode ELM-suppression (RMP) phases; segments of ∼100 ms are cut around D_α-active intervals. All signals are aligned to a common 10 ms grid. Each row provides BES [48] (9 channels), ECEI [49] (4 channels), Mirnov coils (2 channels), time, and the two CES targets T_i and V_{rot} [1]. On this grid T_i is missing in ≈8% of rows and V_{rot} in ≈24%, independently (Fig. 1).

3.2 Sample construction

A training sample at target index t with history window $W{=}4$ consists of `bes` $(W, 9)$, `ecei` $(W, 4)$, `mc` $(W, 2)$, `time_features` $(W, 4)$, `ces_history` $(W, 4)$, the normalized target $[T_i, V_{\text{rot}}] \in \mathbb{R}^2$, and a per-target observation mask $m \in \{0, 1\}^2$. The four time features—lookback seconds, inter-row delta seconds, and their $\log(1{+}x)$ transforms—expose the irregular observation pattern to the model explicitly: the reliability of a past CES value depends strongly on whether it is 10 ms or 200 ms old. The four CES-history channels are the previous normalized T_i, previous normalized V_{rot}, and one observation flag per target (the targets go missing independently, so observation must be tracked per target).

3.3 Design principles

No fabricated labels. We never impute targets to create training rows. A row is kept iff its diagnostic inputs are complete and *at least one* CES target is genuinely observed.

Per-target masked loss. The loss is masked per target, $\mathcal{L} = \sum_{k\in\{T_i,V_{\text{rot}}\}} m_k\,(\hat{y}_k - y_k)^2 / \sum_k m_k$, so a row with only one observed target still supervises that target. The previous both-targets-required filter silently discarded ≈28% of labeled rows; removing it is a pure data gain.

Leakage hardening. (i) *File-level splits*: adjacent rows within a shot are strongly autocorrelated, so splits are made at the shot-file level and pinned to disk; (ii) *train-file-only normalization*: all z-score statistics (NaN-aware for the sparse targets) are estimated on training files only; (iii) *target-step masking*: in `ces_history`, the target timestep's values *and* observation flags are zeroed, so the model cannot read its own answer.

3.4 Data-quality audit: held (forward-filled) V_{rot} values

A late audit found that **54% of observed V_{rot} values are instrument holds**: bit-identical repeats of the previous observation within a contiguous time block (runs up to 1,214 rows; 499/641 files affected). These arise because the native V_{rot} cadence is slower than the row cadence; they are not independent measurements. T_i is unaffected (0.0%). Holds are *excluded from evaluation*: reproducing a forward-fill is not a measurement, and any arm that carries the last value forward—persistence above all—scores it for free. The correction inflates reported physical V_{rot} RMSE by 35–55% (from 22–33 to 35–46 in raw units) and cuts the scored V_{rot} population by roughly 60%. Every headline number in this paper is the corrected one; because the historical convention scored holds, we carry the hold-inclusive population alongside as a sensitivity column (Table 2) and note where the two disagree. They agree on both conclusions.

Holds also contaminate *training*. An initial single-seed A/B suggested they did not—that holds acted as a harmless persistence prior—and the headline family of §6.2 was trained under that assumption. A four-seed paired retrain overturns it. Removing holds at train time (from the supervision targets, from the `ces_history` values *and* observation flags, from the normalization statistics, and from the interpolation anchors), with the file-level split pinned to the control manifest and the scored population matched row-for-row, improves genuine-measurement V_{rot} on *all four* splits: mean paired skill +0.039 (≈4% MSE reduction from the data fix alone), three of four shot-clustered CIs excluding zero, none against. T_i is unaffected (mean +0.004; one split significantly better, none worse). The forward-fill padding was actively teaching the model that copying history is near-optimal, so V_{rot}'s reliance on its own past is *both* real physics and partly an instrument artifact; the honest statement quantifies both.

The window sweep (§6.7) reproduces the effect along an axis it was never tested on: the gain from removing holds decays monotonically with window length (+0.088, +0.048, +0.035, +0.003, +0.006 at $W = 2, 3, 4, 6, 8$). With holds present, a short window's single history slot is often a forward-filled copy carrying no information, while a long window still reaches a genuine reading— so holds were *penalising short windows*, not rewarding long ones. Uncorrected, this artifact alone manufactures an apparent "V_{rot} needs a long history" trend (+0.118 → +0.202) that vanishes under the correction.

We therefore adopt hold-free training as the convention for all subsequent runs. Its effect on the headline is a strict improvement that changes no verdict: retrained hold-free, T_i still PASSes on 4/4 splits (+0.184, +0.291, +0.238, +0.221) and every V_{rot} point estimate rises (+0.203 → +0.226, +0.162 → +0.202, +0.100 → +0.143, +0.183 → +0.210) while remaining n.s. on 3/4. Both conclusions of this paper are thus invariant to the data treatment, which is the reason we report the pre-registered family as the headline and this one as its replication.

4 Model

The predictor is a compact **201,258-parameter** late-fusion network; the bottleneck in this problem is information, not capacity (§6.7).

Per-diagnostic encoders. Each modality (BES, ECEI, MC) is encoded by its own time-aware 1-D CNN that consumes the modality jointly with the time features and history (two Conv–BatchNorm–GELU blocks, global average pooling, linear projection to 96-d), preserving each diagnostic's channel structure until fusion; a fourth, smaller CNN encodes the time features alone. Convolution is preferred to recurrence *on the sensor streams* because gap-removal makes their effective sampling irregular, violating the uniform-step assumption RNNs rely on.

History encoder: observation-masked attention pooling. The CES history, concatenated with the time features, is read by a single-layer *bidirectional* GRU (hidden 64), and its per-timestep outputs are pooled by *two independent multi-head additive-attention readouts*, one routed to the T_i head and one to the V_{rot} head. The window is strictly past-only—the target row is the *last* row of the window (§3)—so bidirectionality runs over history only and grants no future access; it lets the readout weigh each history row in the context of the rows that follow it inside the window.

The one non-generic ingredient is the pooling mask. Before each target's softmax, every timestep whose *per-target* observation flag is zero—the fully masked target row, and any history row where that particular target was not observed—is driven to $-\infty$, so attention mass can land only on rows where that target was genuinely measured. The two targets are masked independently, because they go missing independently; a target with no observed row in its window falls back to attending over the whole window, which prevents an all-($-\infty$) softmax.

This is the inductive bias of interpolation, written into attention. PCHIP and linear interpolation are strong precisely because they use *only* the observed samples flanking a gap and never the gap itself. An unmasked attention pool can spend weight on the zeroed target row and on unobserved history rows whose

GRU states carry no measurement, diluting the readout and forcing the network to learn that suppression implicitly through the flag channels. Hard-masking supplies the same bias directly and at *zero parameter cost*—the change is purely *which* timesteps each pool may weight—and it was the last change accepted by the selection protocol of §7.

Physics-informed per-target routing. The T_i head fuses fast diagnostics + time + history; the V_{rot} head fuses *history and time only*, deliberately excluding the fast diagnostics, which ablations (§6.6) show carry no usable rotation information at 10 ms. Each pooled history summary is normalized and projected by its own target-specific LayerNorm–Linear–GELU block before entering that target's MLP head, and the two heads are sized to their inputs (72k parameters for T_i, 14k for V_{rot}). Outputs remain in normalized units; denormalization happens only in evaluation.

Search lessons: capacity is not where the remaining gain is. Across ~40 controlled architecture iterations spanning two search rounds, normalization placement was the stability turning point of the earlier round (validation loss $0.89 \to 0.49$); attention pooling was the most reliable single improvement and is the one mechanism that survives into the final model; and aggressive capacity scaling, complex skip topologies, and extra convolutional extractors consistently failed to improve held-out performance. Together with §6.7 this negative carries information of its own: it says that further capacity engineering is the wrong place to spend, and it is what lets us say with some confidence, in §9, *where* the next gain actually is.

5 Evaluation methodology

Metric. All errors are denormalized to physical CES units and computed per target. The primary score is Murphy's skill [14] against the pre-registered headline baseline, $\mathrm{skill}_{\mathrm{PCHIP}} = 1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{PCHIP}}$ (positive = model better; 0 = tie). Every arm (model and all baselines) is scored on the *identical* (file, row) sample set under the same per-target keep mask. Interpolants exclude the target's own value (no leakage). Each shot file contains multiple measurement segments separated by gaps ≥ 0.5 s (all 641 shots; median 2 per file, median gap 6.3 s); neither the interpolants nor the model's input window bridges a segment boundary, and where a neighbor beyond the boundary would be needed the interpolant predicts the persistence value instead (no population shrinkage; PR2).

Three-way split; selection never sees TEST. Files are split train/val/test; the test set was reserved *before* any architecture search and is never read during model selection, so headline numbers carry no winner's-curse bias. The canonical (seed-42) test set contains 33,693 scored samples: $n = 32{,}787$ observed T_i across 96 shots, and $n = 10{,}729$ genuine V_{rot} across 61 shots. The V_{rot} count is small because holds are excluded (§3.4); scoring holds as if they were measurements would inflate it to 27,437, and we report that population only as a sensitivity check. Every number in this paper is read from frozen run directories by a single collection script, so the text, the tables, and the figures cannot drift apart.

Pre-registration. Four rules were fixed in writing before viewing test numbers: **PR1** the headline comparator is PCHIP (chosen a priori for monotonicity/no ringing at transients; the full ladder {persistence, linear, PCHIP, AR} is also reported); **PR2** interpolation predicts everywhere the model is scored, with persistence fallback where no future neighbor exists (no population thinning); **PR3** a minimum test-set floor ($\geq$15 shots, $\geq$3,000 observed T_i samples; met); **PR4** significance means a shot-clustered bootstrap 95% CI on the skill excluding zero.

Shot-clustered paired bootstrap. Adjacent CES rows within a discharge are strongly correlated; treating samples as independent would grossly understate uncertainty. Per-sample paired errors $\mathrm{SE}_{\mathrm{model}} - \mathrm{SE}_{\mathrm{PCHIP}}$ are aggregated by shot, and whole shots are resampled with replacement (10,000 resamples, fixed seed). The resulting CI reflects genuine shot-to-shot generalization; its cost is that the effective sample size is the number of shots ($\approx$96), which bounds statistical power throughout.

Table 1: Physical-unit RMSE ladder, final model, canonical held-out test split (seed 42), genuine-measurement evaluation. Lower is better. The two interpolants are the only arms that read the future. V_{rot} RMSE is ≈35 rather than the ≈23 that a hold-inclusive population reports, because held targets carry near-zero baseline error (§3.4).

Arm	Information access	T_i RMSE	V_{rot} RMSE
Model (nowcaster)	fast diag. + past CES	**407.0**	**35.0**
Linear interpolation	past + future CES	441.0	38.4
PCHIP interpolation	past + future CES	449.3	39.2
Persistence	last observed CES	504.3	44.4
AR (local, causal)	past CES only	2425.9	91.5

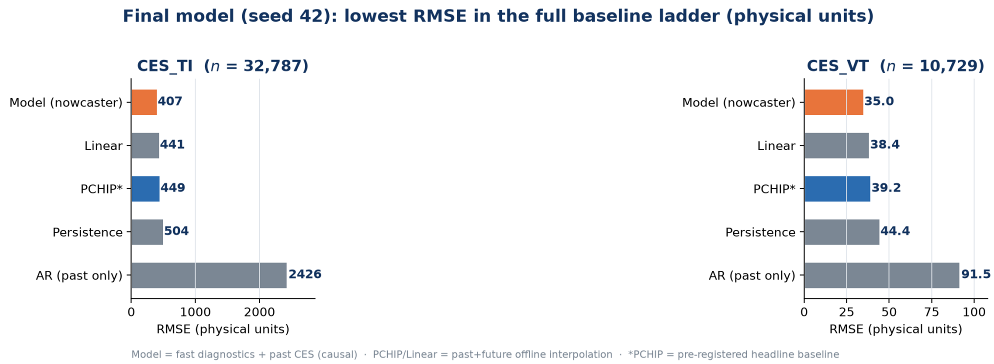

Figure 2: RMSE ladder for the final model (seed 42, held-out test, genuine measurements): the causal model attains the lowest RMSE for both targets, decisively beating the causal baselines and edging the future-using interpolants. Same numbers as Table 1; both are generated from the same frozen artifacts.

The MNAR problem, and what we do about it. Truly missing points have no ground truth, so skill is measured by masking *observed* points. Because CES missingness is MNAR (drop-out at low S/N, ELMs, transitions), observed-point skill is an optimistic estimate of performance at genuinely missing points. Stating that and stopping there leaves the paper's central number with an unbounded correction attached to it, so we do not stop there: §6.4 reweights the scored points onto the covariate distribution of the *actually missing* ones and reports how much of the skill survives, together with what fraction of the missing population this model can be applied to at all.

6 Results

6.1 The model dominates every causal baseline

Table 1 shows the physical-unit RMSE ladder on the canonical test split. The model has the lowest RMSE for both targets and beats the causal baselines—persistence and past-only AR, i.e. everything actually available in real time—by a wide margin. This is the most robust result in the paper: for any online setting, where future CES is by definition unavailable, the model is the clear winner.

6.2 Headline: T_i beats future-using interpolation on four independent splits

With the final architecture (bidirectional-GRU history encoder + per-target observation-masked multi-head attention, W=4; selected by the protocol of §7 on validation skill only), we evaluated four *independent* held-out test splits (seeds 42, 1, 7, 123; the latter three were never used in any selection step). Table 2 reports both evaluation populations.

- T_i**: PASS on all four splits, under both evaluation populations.** $\mathrm{skill}_{\mathrm{PCHIP}} = +0.18$ to $+0.28$

Table 2: Held-out test $\text{skill}_{\text{PCHIP}}$ with shot-clustered 95% CIs (10,000 resamples). *genuine* excludes instrument holds from scoring (headline; §3.4); *holds kept* is the sensitivity check on the historical population. Seeds 1, 7 and 123 were never touched by model selection.

		genuine measurements only		holds kept	
Target	seed	$\text{skill}_{\text{PCHIP}}$ (95% CI)	gate	$\text{skill}_{\text{PCHIP}}$	gate
T_i	42	$+0.179$ $[+0.007, +0.302]$	PASS	$+0.257$	PASS
	1	$+0.197$ $[+0.091, +0.276]$	PASS	$+0.194$	PASS
	7	$+0.280$ $[+0.192, +0.347]$	PASS	$+0.263$	PASS
	123	$+0.263$ $[+0.109, +0.348]$	PASS	$+0.280$	PASS
V_{rot}	42	$+0.203$ $[-0.225, +0.362]$	n.s.	$+0.154$	n.s.
	1	$+0.162$ $[+0.021, +0.257]$	PASS	$+0.109$	n.s.
	7	$+0.100$ $[-0.586, +0.291]$	n.s.	$+0.065$	n.s.
	123	$+0.183$ $[-0.151, +0.265]$	n.s.	$+0.127$	n.s.

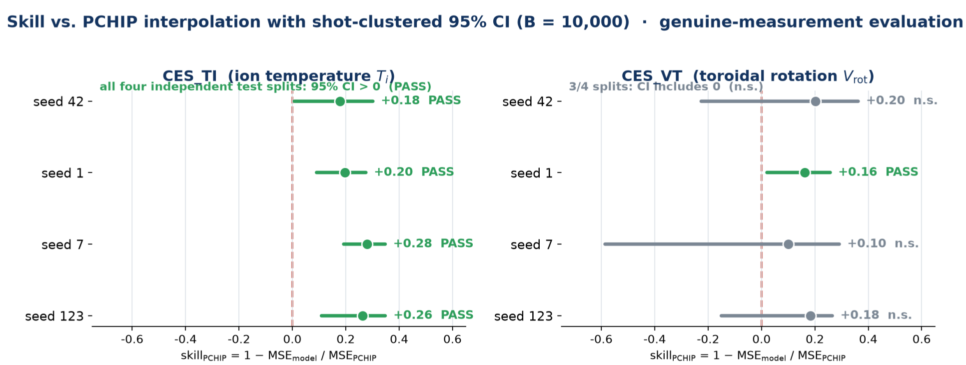

Figure 3: Forest plot of $\text{skill}_{\text{PCHIP}}$ with shot-clustered 95% CIs across four independent held-out splits. T_i excludes zero on all four; V_{rot} on none.

genuine (+0.19 to +0.28 with holds kept); every shot-clustered CI excludes zero. Against the *stronger* linear interpolant the margin narrows but survives on 3/4 splits genuine (+0.148 n.s. on seed 42; +0.167, +0.259, +0.234 PASS) and 4/4 with holds kept. We flag the seed-42 linear result rather than quoting only the pre-registered PCHIP comparator.

- **V_{rot}: a tie, not a win.** Point estimates are positive on every split under both populations, but only seed 1 clears PR4 (genuine +0.162 $[+0.021, +0.257]$; 0/4 with holds kept). One significant split out of four is what we would expect from noise, so we claim no rotation win.
- **Post-hoc check: the strongest offline arm is a GP, and the model ties it.** Outside the pre-registered ladder we added the fusion community's standard profile-fitting tool, GP regression, as a post-hoc arm on the byte-identical population (exact Matérn-3/2 + white noise, local nearest-16+16 fit, per-sample marginal-likelihood grid selection — deterministic; it never reads the target's own value and inherits the PR2 fallback). The GP turns out to be the strongest offline smoother, beating PCHIP significantly on all four T_i splits (+0.21–+0.28) — the mechanism by which linear beat PCHIP taken one step further: instead of passing exactly through noisy anchors it averages the noise out — and the model *ties* it (significantly better on 1/4, worse on 0/4, point estimates −0.08 to +0.09; V_{rot} all n.s.). The observed-population claim is therefore "beats the pre-registered interpolants and matches a future-using, per-sample-tuned smoother"; the deployment claim is untouched, since a GP without future anchors does not exist online. Named measurements that could resolve the tie: more test shots (seed 7 already resolves it), and a *causal* past-only GP arm, not yet run.

An honest progression note: the initial baseline model was *not* significant (single split, +0.088, 95% CI $[-0.221, +0.323]$). Switching the architecture-search selection gate from validation loss to the clean

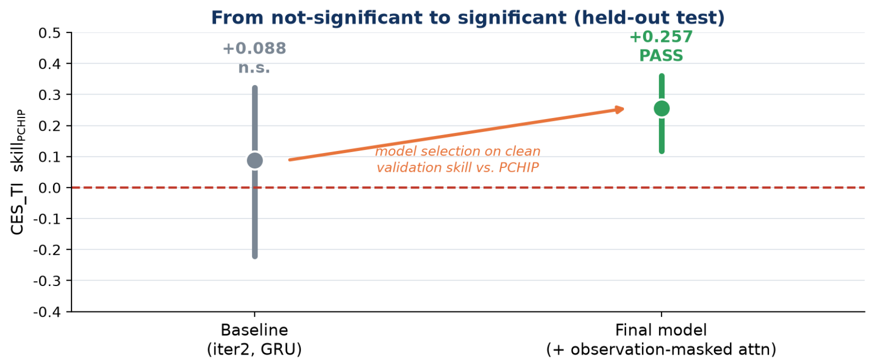

Figure 4: From non-significant (baseline, +0.088 n.s.) to significant (final model, +0.18–+0.28, PASS ×4): the decisive change was selecting architectures on clean skill rather than validation loss.

Table 3: Gap-stratified skill, four test splits pooled, shot-clustered 95% CIs (genuine measurements). **PCHIP reads a future anchor; persistence does not.** Bold = CI excludes zero.

Δt	n	shots	vs. PCHIP (future-using)	vs. persistence (causal)
T_i				
≤ 15 ms	134,629	298	+**0.262** [+0.20, +0.31]	+**0.407** [+0.36, +0.45]
$(15, 25]$	2,987	259	+**0.183** [+0.05, +0.30]	+**0.384** [+0.28, +0.48]
$(25, 35]$	784	122	+**0.402** [+0.21, +0.57]	+**0.564** [+0.41, +0.68]
$(35, 55]$	395	98	+0.143 [−0.21, +0.41]	+0.169 [−0.15, +0.45]
$(55, 105]$	163	57	+**0.282** [+0.03, +0.49]	+**0.287** [+0.04, +0.51]
> 105 ms	167	62	−**0.542** [−2.07, −0.08]	+0.266 [−0.00, +0.46]
all > 15 ms	4,496	272	+**0.191** [+0.10, +0.28]	+**0.388** [+0.32, +0.46]
all > 45 ms	435	105	−0.057 [−0.45, +0.21]	+**0.271** [+0.12, +0.41]
$V_{\rm rot}$				
≤ 15 ms	51,457	195	+0.209 [−0.02, +0.29]	+**0.368** [+0.17, +0.45]
all > 15 ms	1,756	161	+0.027 [−0.20, +0.24]	+**0.309** [+0.13, +0.47]

interpolation-relative skill produced the final model, which is significant on all four splits (Fig. 4). Both stages are scored on the same hold-inclusive population, because that is the only one on which the earlier checkpoint was ever evaluated; comparing the two across different populations would not be a comparison. We report both stages.

6.3 Where the skill lives, and who the right opponent is at large gaps

Stratifying by Δt, the time since the last observed CES value, ≥96% of real targets lie in the small-gap regime ($\Delta t \leq 15$ ms). Per-seed the wide bins hold only tens of samples, which is why earlier readings of them were not load-bearing; we therefore *pool the four test splits* and cluster the bootstrap on the physical discharge, so a shot appearing in more than one split counts once. That buys enough shots to put a CI on the wide bins for the first time (Table 3).

It also forces a question the PCHIP comparison cannot answer. As Δt grows, PCHIP's problem gets *easier*—it interpolates between two real observations that straddle the gap—while ours gets harder, since we extrapolate from the past alone. More to the point, at run time there is no future anchor, so PCHIP is not an option at all. The baseline a deployed nowcaster must actually beat at large gaps is the best causal one: persistence. We therefore report both.

Three readings, in decreasing order of confidence.

The T_i win is not confined to adjacent history. Pooling makes the non-adjacent regime measurable, and the model beats future-using PCHIP there too: +0.191 [+0.10, +0.28] over all $\Delta t > 15$ ms, with the

Table 4: Skill reweighted from the observed population onto the genuinely-missing, in-domain one. Shot-clustered 95% CIs (2,000 resamples; the interval covers error variability at fixed stratum weights).

Target	seed	observed-weighted	missing-matched (95% CI)	optimism
T_i vs. persistence	42	+0.349	+**0.292** [+0.162, +0.415]	+0.056
	1	+0.380	+**0.308** [+0.143, +0.429]	+0.072
	7	+0.405	+**0.293** [+0.192, +0.441]	+0.112
	123	+0.415	+**0.293** [+0.155, +0.392]	+0.121
T_i vs. PCHIP	42	+0.146	+0.061 [−0.075, +0.267]	+0.084
	1	+0.188	+0.132 [−0.064, +0.274]	+0.056
	7	+0.280	+**0.211** [+0.050, +0.374]	+0.069
	123	+0.264	+0.175 [−0.025, +0.290]	+0.090

individual $(15, 25]$, $(25, 35]$ and $(55, 105]$ bins each significant on their own. This is a stronger claim than the paper could previously make, and it is the regime where gap-filling is least trivial.

At the widest gaps, interpolation genuinely wins—but only because it can see the future. Beyond 105 ms the model is significantly *worse* than PCHIP (−0.542, CI entirely below zero). We report this plainly. Against the causal reference in the same bin it is a tie (+0.266, lower bound at zero), and pooled over all $\Delta t > 45$ ms it *beats* persistence significantly (+0.271 [+0.12, +0.41]) while merely tying PCHIP (−0.057, n.s.). So the correct statement is not "the model fails at large gaps" but "the large-gap regime belongs to two-sided interpolation, and two-sided interpolation is exactly what a real-time system does not have."

V_{rot} ties interpolation but beats every causal alternative. The global V_{rot} tie of §6.2 is a tie *with future-using methods*. Against persistence, V_{rot} is significantly better at small gaps (+0.368) and stays significantly better beyond 15 ms (+0.309). For an online virtual sensor this is the operative comparison, and by it the rotation channel is useful even though the headline calls it a tie.

6.4 How much of this survives at the points that are actually missing?

Everything above is measured on CES points that happen to be *observed*. The points a nowcaster exists to fill are missing, and CES missingness is not random, so the two populations differ. We can never measure error where there is no ground truth—but we can measure how skill varies over covariates that are computable at observed *and* missing rows, and reweight. Two covariates carry the mechanism: Δt since the last genuine observation (a missing point sits deeper into a drop-out), and the input-only local-activity flag of §6.9 (the ELM/transition axis), which is defined at a missing row precisely because it is computed from a neighbour set that excludes the row itself. We post-stratify on $\Delta t \times$ activity, drop any stratum with fewer than 30 scored samples rather than trust it, and report the resulting coverage next to every estimate. The assumption—that within a stratum missing and observed rows are exchangeable—is stated, not assumed away. As a control, the activity rate computed on this grid agrees with the flag stored in the frozen artifacts to within 0.004 (T_i) and 0.009 (V_{rot}), so the two sides of the reweighting use the same variable.

First, a scope result we had not previously stated. A sample is only predictable at all if an observed value of that target lies *inside* the model's window. Of the genuinely missing T_i rows, **54.1% are in-domain** in this sense; of the genuinely missing V_{rot} rows, **only 4.8% are**. So at W=4 this method addresses about half the real T_i gap problem and a small corner of the V_{rot} one—not because it predicts them badly, but because it cannot be applied. That is a coverage limit, it is fixable by construction (§9), and it re-opens the window question of §6.7: W is irrelevant to skill but decisive for how much of the actual missingness is reachable.

Second, the optimism itself is small—for the causal comparison. Restricted to in-domain missing points (95–100% stratum coverage):

Table 5: Temporal (campaign) split, four initialisations. The test period is harder for every arm (PCHIP RMSE 601 vs. 449 on the random split), but skill normalises that away, so these are losses *relative to* interpolation.

	T_i		V_{rot}	
init seed	vs. PCHIP	vs. persistence (95% CI)	vs. PCHIP	vs. persistence (95% CI)
42	$+0.051$	$+\mathbf{0.275}$ $[+0.175, +0.342]$	-0.061	$+\mathbf{0.223}$ $[+0.016, +0.316]$
1	$+0.044$	$+\mathbf{0.270}$ $[+0.153, +0.343]$	-0.005	$+\mathbf{0.264}$ $[+0.099, +0.356]$
7	-0.148	$+0.123$ $[-0.061, +0.241]$	-0.156	$+\mathbf{0.153}$ $[+0.019, +0.242]$
123	-0.018	$+\mathbf{0.222}$ $[+0.099, +0.303]$	-0.056	$+\mathbf{0.226}$ $[+0.025, +0.355]$
verdict	0/4 PASS	**3/4 PASS**, mean $+0.222$	0/4 PASS	**4/4 PASS**, mean $+0.216$

Table 6: The two claims under stress. Only one of them is robust, and it is the one that matters for deployment.

Evaluation	vs. PCHIP (offline, future-using)	vs. persistence (causal)
Random split, observed points	$+\mathbf{0.18}$ to $+\mathbf{0.28}$, **4/4**	$+0.35$ to $+0.42$
Reweighted to missing points (§6.4)	$+0.06$ to $+0.21$, 1/4	$+\mathbf{0.29}$, **4/4**
Temporal campaign split (§6.5)	-0.15 to $+0.05$, 0/4	$+0.12$ to $+0.28$

The two comparisons come apart, and the split is the most useful thing in this section. Against **persistence—the baseline an online system actually runs—the T_i advantage survives reweighting on all four splits** at $+0.29$, with a remarkably stable point estimate and every CI excluding zero; the MNAR correction costs only 0.06–0.12. Against **future-using PCHIP the advantage does not survive**: 1/4 splits significant after reweighting. This is consistent with §6.3—missing points sit at larger Δt, which is exactly where a two-sided anchor helps most and a causal extrapolator least.

We therefore state the claim the evidence supports at the points that matter: *at genuinely missing, in-domain timesteps, the nowcaster is significantly better than any causal CES-only method, and is not shown to beat offline interpolation.* The offline comparison remains the harder, more informative bar for the observed population, and we keep it as the headline for that population; it is not the claim we make about deployment. V_{rot} behaves the same way but on a much thinner base (2/4 significant vs. persistence, and only 4.8% of its missing rows in-domain), so we draw no deployment conclusion for rotation.

6.5 Does it survive a campaign shift?

The second thing deployment does that a random split does not is move forward in time. We therefore order all 641 discharges by shot number (30801–32751, contiguous) and cut strictly in time: **train 416 shots [30801, 31991], val 128 [32002, 32310], test 97 [32312, 32751]**, with no test shot preceding any train shot. Everything else is pinned to the headline protocol, so the only changed variable is the split rule. The split is fixed, so the four runs vary the *initialisation* seed only—replication here is over initialisation, not over splits, and we say so rather than presenting it as "four seeds".

Two things in that table are worth stating explicitly. First, the interpolation advantage does not merely shrink—it is gone, against PCHIP *and* against the stronger linear interpolant, on 0/4 initialisations. Second, V_{rot}—the target we report as a tie—beats persistence with a CI excluding zero on **4/4** initialisations *across a campaign boundary*. Rotation is not a failed target; it is a target whose only informative comparison is the causal one.

Two independent stress tests give the same answer. Table 6 puts them side by side. The advantage over *future-using interpolation* is a property of the observed population of a random split and survives neither stress test. The advantage over *the baselines a real-time system can actually run* survives both.

Table 7: Input-modality ablation (validation split, skill vs. persistence, physical units). Fast diagnostics alone carry genuine T_i information but are catastrophically uninformative for V_{rot}.

Model inputs	T_i skill	V_{rot} skill
Full (history + fast + time)	+0.43	+0.30
History only (*no fast*)	+0.46	+0.29
Fast diagnostics only (*no history*)	+0.37	−0.64

Why it degrades—measured, not asserted. The pattern implies a specific mechanism: what fails to transfer should be the *fast-diagnostic* pathway, since that is exactly what buys the edge over CES-only interpolation, while the CES-history pathway sees the same physical quantity in the same units in both periods. Measuring the train→test drift per channel, in the units the model was normalized in, confirms it: the median location shift is **1.22 σ for BES** and **0.53 σ for ECEI**, with scale ratios of 0.75 and 0.62—against **0.115 σ and a scale ratio of** 1.06 **for the CES targets themselves**. The fast diagnostics drift 5–11× more than the quantity being predicted. The encoder is being fed inputs more than a standard deviation outside the range it was normalized on, while the history pathway is untouched, which is why the margin over persistence largely holds.

This also indicts one of our own design choices, in a useful way. Normalizing with *train-file-only* statistics (§3) is the correct leakage-hardening choice under a random split and is precisely what breaks under campaign shift. The targeted repair is per-shot or causal-running standardization of the fast diagnostics: it uses only that shot's own data, so it remains leak-free and is available at run time. We flagged one honest caveat rather than presenting it as a certain win—per-shot standardization also discards absolute level, which may itself carry T_i information (higher BES ↔ higher density), so it may trade random-split skill for transfer. That trade-off is a controlled experiment, not an assumption, and we then ran it.

The repair works, and the trade-off is small but real. Standardizing each discharge's fast diagnostics by *its own* mean and variance at load time—targets untouched, so the physical-unit inverse transform and every baseline are unchanged—is the single controlled variable against the runs above. On the campaign split it recovers most of what was lost: paired T_i skill improves on **4/4 initialisations by** +0.10 **to** +0.29 **(mean** +0.155**), every 95% CI excluding zero**, and the vs-PCHIP gate goes from **0/4 to 2/4** (+0.18, +0.14, +0.19, +0.08), where the unrepaired arm had been *below* interpolation on two of four. V_{rot} is unmoved (mean −0.019), which is the expected negative control: the V_{rot} head is blocked from the fast diagnostics by construction, so standardizing them cannot reach it. On the random (headline) split, paired against the matched control, the arm shows **no significant loss on any seed** (mean −0.036; the worst seed is −0.127 with a CI upper bound of +0.008), so the honest statement is "no measurable cost", not "free". This also settles the mechanism causally: if the campaign loss were the physics changing between periods, removing an input *level* shift could not have recovered it. The remaining gap to deployment is that an online estimator cannot see a discharge's future to compute its variance, so per-shot standardization is the offline upper bound of this family and an expanding-window or EWMA estimator is the deployable version—which we have not measured.

6.6 The $T_i \leftrightarrow V_{\mathrm{rot}}$ information asymmetry

A controlled input-modality ablation (identical training budget; one modality group zeroed at a time; persistence always computed from real history) explains *why* the two targets behave differently (Table 7, Fig. 5).

- T_i**: fast diagnostics carry real information.** Fast-only still beats persistence by a wide margin (+0.37, i.e. most of the full model's +0.43). The physical channel is collisional electron–ion coupling ($t_{ei} \propto T_e^{3/2}/n_e$): ECEI supplies T_e and BES supplies n_e structure.
- V_{rot}**: information is almost entirely CES history.** Fast-only scores −0.64, with $R^2 = -0.06$ against the mean predictor—that is, strictly worse than ignoring the inputs and predicting the mean.

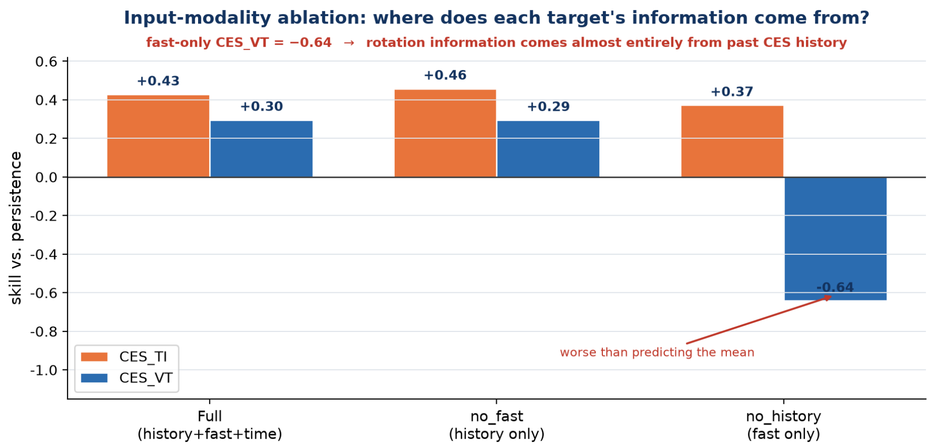

Figure 5: Modality ablation: fast-only beats persistence for T_i but is far worse than the mean for V_{rot} — the information asymmetry.

Table 8: Window sweep, mean held-out test skill$_{\text{PCHIP}}$ over 4 seeds (PASS = number of seeds whose shot-clustered 95% CI excludes zero). History-0 is the `no_history` ablation at W=4, which zeroes the CES-history tensor while leaving the sensor window intact; W=1 is not constructible.

History obs.	W	T_i skill	PASS	V_{rot} skill	PASS
0	4	−0.026	0/4	−0.783	0/4
1	2	+0.238	**4/4**	**+0.206**	0/4
2	3	**+0.246**	**4/4**	+0.203	1/4
3	4 *(old default)*	+0.221	3/4	+0.190	1/4
5	6	+0.190	3/4	+0.205	1/4
7	8	+0.216	**4/4**	+0.204	2/4

Toroidal rotation is governed by momentum sources that none of our inputs observe—the external NBI torque and intrinsic rotation drives [50]—and the Mirnov signals are sampled at 100 Hz, aliasing away the kHz mode-rotation frequencies that could otherwise proxy rotation.

This asymmetry was predicted from physics before the ablation and is confirmed by it; the V_{rot} non-win is a finding about diagnostic information content, not a model failure. It also motivates the per-target routing of §4.

6.7 How much history is needed? One observation

The history window was originally an unexamined default ($W = 4$). We replace the assertion with a curve and a stated selection rule: **24 independent runs**, $W \in \{2, 3, 4, 6, 8\}\times$ seeds $\{42, 1, 7, 123\}$, plus a *history-0* point $\times 4$ seeds, each scored on its own held-out test split with the same protocol as the headline. Three controls make the points comparable, each verified on the data rather than assumed: the file-level split is window-invariant (all 24 runs evaluate the same 96 test shots per seed); a new per-shot sample cap prevents temporal-subset augmentation—which explodes from 240k samples at W=2 to 30.1M at W=8—from letting a few long-block discharges dominate the training subset more strongly the larger W is; and the scored population shrinks only 1.8% from W=2 to W=8. Training is hold-free throughout (§3.4), which is essential here: a hold is a literal copy of the previous reading, so under contamination a longer window merely supplies more copies of the same number. All runs use the final architecture.

History is essential, and its first observation carries essentially all of it. Removing previous-CES entirely puts T_i *below* PCHIP (−0.026, 0/4) and V_{rot} at −0.78, i.e. ≈1.8× PCHIP's MSE. So the fast

Table 9: Complexity ladder: held-out test $\text{skill}_{\text{PCHIP}}$, hold-free training, genuine-measurement evaluation. The last column is the fraction of the persistence→full-model gap that the interpretable model recovers.

Arm	params	seed 42	1	7	123	gap recovered
T_i						
Persistence	0	−0.260	−0.320	−0.251	−0.256	—
Anchor+Δ	1,258	−0.107	−0.138	−0.097	−0.109	**31.5%**
Full model	201,258	+0.238	+0.184	+0.291	+0.221	100%
V_{rot}						
Persistence	0	−0.283	−0.297	−0.287	−0.307	—
Anchor+Δ	1,258	−0.253	−0.269	−0.241	−0.278	**7.0%**
Full model	201,258	+0.227	+0.202	+0.143	+0.210	100%

diagnostics alone do not reach interpolation parity: the model's entire margin comes from combining them with past CES. Yet a *single* past observation lifts both targets to their maximum at once (T_i +0.238 4/4, V_{rot} +0.206), and the curve is flat from there on—T_i spans 0.190–0.246 and V_{rot} 0.190–0.206 (range 0.016), while the seed spread *within* a single point is 0.07–0.16, wider than the whole curve. This is the sharpest statement of the information-limited reading: neither more history nor (§4) a more expressive encoder moves the result, but deleting one observation destroys it.

Selection rule and its answer. Applying "smallest W that reaches plateau" returns $\mathbf{W = 2}$; both targets are already at plateau there, and the incumbent $W = 4$ is *below* the W=2/W=3 values on both targets. Nothing in the data supports paying for a longer window on accuracy grounds. The one defensible reason to go wider is **coverage, not skill**: a sample is scored only when an observed value exists inside the window, so W=2 silently drops targets whose adjacent row lacks that reading. Total scored counts barely move (V_{rot} 52.1k → 53.2k), but the hard subset grows sharply—$\Delta t > 15$ ms goes 456 → 1,958 and $\Delta t > 45$ ms goes 14 → 135 from W=2 to W=8. A wider window does not predict long-gap samples *better*; it predicts 4–10× *more of them at all.*

Honest caveats. These are *unpaired* independent runs, so every difference along the curve sits inside seed noise: the claim is the *absence* of a window effect, not a ranking among windows. And because of the per-shot cap the absolute skills are not directly comparable to the headline family of §6.2, which had no such cap; the sweep is designed for *within-curve* comparison, where every point shares it.

6.8 What does the complexity buy? A ladder down to 1,258 parameters

A fair criticism of this model is that it is a sedimented search result rather than a structure derived from the task, and that its 0.2M parameters are unexplainable. The useful reply is not a small model that also happens to work—it is to *measure what the complexity earns.* We therefore built the most interpretable predictor the task admits and put it on the same bar.

The **anchor+Δ** model predicts each target as a sum of three named terms: a persistence-style *anchor* (the nearest observed CES value, blended toward the observed mean by a learned smoothing weight), a *slope* term (the slope of the last two observations × the gap), and a *rate* term per diagnostic (BES/ECEI/MC summary statistics → a rate of change × the gap). It has **1,258 parameters**, 0.6% of the full model. Every learned term is zero-initialised, so training *starts exactly at persistence* and the skill it reaches is literally what learning added; a `decompose()` call returns each term's contribution, so any prediction can be read as "anchor plus this much slope plus this much BES-driven rate". It was trained and scored under the identical protocol, hold-free, on the same four splits.

Three things follow, and the consistency across four independent splits (28–36% for T_i) is what makes them sayable.

- **The complexity earns its keep, quantifiably.** A fully interpretable predictor with 0.6% of the parameters recovers about **a third** of the T_i margin over persistence and **7%** of the V_{rot} margin. The rest is what the learned multimodal encoder buys, and a paired shot-clustered comparison rejects the anchor on 4/4 splits.

- **The interpretable rung is not a strawman.** At −0.113 it sits far above persistence (−0.272), so the named-terms form does learn something real—it is simply not enough.
- **The shortfall is informative about the problem, not just the model.** The anchor is *linear in the gap*. That this captures only a third of the recoverable signal says the structure is substantially non-linear and non-local, so a more explainable model needs a different functional form rather than more linear terms; and the V_{rot} figure of 7% says its signal is not a local slope at all.

6.9 The edge concentrates in high-variability neighborhoods

Using a target-independent, input-side proxy for local activity (large neighbor-bracket slope and/or high local CES-neighbor variance, computed *excluding* the target row), we split evaluation into "peak" (high-local-activity) and bulk regions (validation split, shot-clustered CIs; Fig. 6):

- T_i: global +0.27 → peak **+0.70** (CI $[+0.50, +0.85]$, PASS; 4,764 rows over 124 shots). Interpolation is near-optimal on the smooth bulk; the model's value is in the active stretches.
- V_{rot}: global +0.13 → peak **+0.44** (CI $[+0.07, +0.73]$, PASS, robust across three sensitivity settings but with a thin lower bound). The asymmetry is therefore *regional*: globally V_{rot} ties interpolation, but in high-activity neighborhoods — where smooth past+future interpolation is worst — even the history-driven V_{rot} predictor adds value. We report this as a regional strength on the optimism-caveated validation split, not a reversal of the global null result.
- A peak-restricted ablation (zeroing BES/ECEI/MC) hurts T_i significantly (paired-difference CI $\gg 0$) but leaves V_{rot} unchanged: the T_i peak edge genuinely uses the multimodal signal, while the V_{rot} peak edge is history-driven.

Decomposing the V_{rot} result: instrument holds are part of the peak. Because more than half of the *observed* V_{rot} population is forward-filled holds (§3.4), the peak split and the hold pattern could be entangled, so we crossed them directly on the held-out test splits under the hold-inclusive treatment (evaluation only; the hold flag is recomputed with the loader's own rule and aligned row by row). Two things came out, and the first was the opposite of what we expected. **Peaks are hold-*richer* than the bulk on all four splits** (68/62/71/73% versus 58/51/48/46%): a forward-filled staircase—flat, flat, jump—is itself a large local slope, so an activity detector built from CES neighbors partly keys on the instrument's hold pattern. **And the global V_{rot} number is an average over three regimes with different signs.** On genuinely measured, high-activity rows the model reaches +0.55–+0.63 against future-using PCHIP (positive on 4/4, significant on 1/4 at ≈2–3 k rows over ≈63 shots) and +0.75–+0.82 **against the causal baseline, significant on 4/4**. On the genuine quiet bulk it ties interpolation (≈ 0 on all four). On hold rows it loses by −48 to −411—which is structural rather than a model failure, since PCHIP passes exactly through a value that is by construction the previous one, and no causal method can win there. Restricting the scored population to genuine rows lifts every split (+0.154 → +0.201, +0.109 → +0.126, +0.065 → +0.098, +0.127 → +0.170) without reaching the pre-registered gate, so hold dilution is real but is not by itself why V_{rot} is a tie; power is. T_i, which has essentially no holds, is the clean control and reproduces the peak concentration on 4/4 splits (+0.59–+0.68, all significant), so the concentration finding does not depend on the artifact. The measurement that would sharpen this is a hold-free activity detector, which we have not run.

7 Architecture-selection protocol

The final architecture was reached by a sequence of controlled experiments under a fixed protocol. Each candidate was *one controlled change* to the model that had to preserve a hard data contract (interfaces, masking, and leakage rules); it was smoke-validated, trained to completion, and then scored on the *clean, non-augmented* validation skill$_{\mathrm{PCHIP}}$—never the augmented validation loss, and never any test split. A

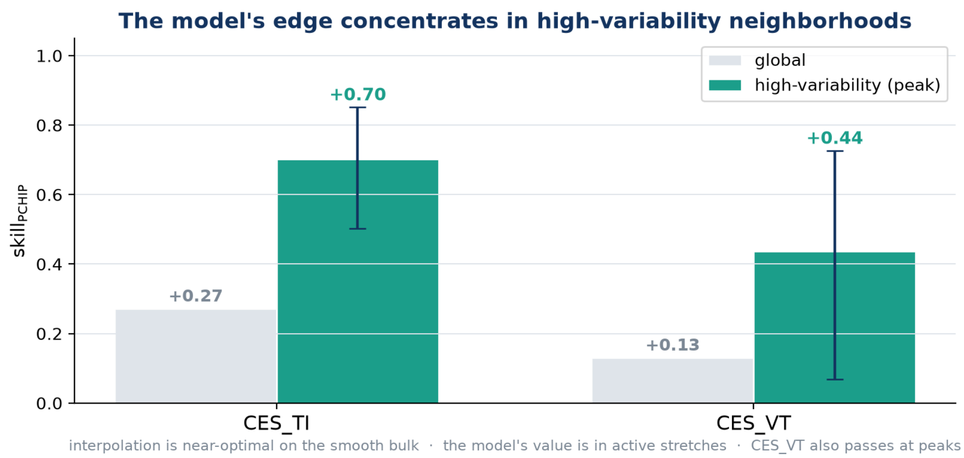

Figure 6: Global vs. high-variability ("peak") skill with shot-clustered 95% CIs. The model's advantage over interpolation concentrates where interpolation is weakest; V_{rot} passes at peaks despite the global tie.

candidate was **kept** iff it improved the best score so far, and otherwise **discarded**, with `model.py` rolled back to the best snapshot; training failures were treated as repair signals rather than as evidence about architecture quality. Selection therefore never builds on a regression.

The choice of gate matters more than it may appear. The augmented validation loss is not the quantity the paper reports: temporal-subset augmentation replicates easy, densely observed neighborhoods, and minimizing squared error on that distribution rewards smoothing—exactly what future-using interpolation already does well. Scoring candidates directly on the clean interpolation-relative skill aligns the selection gate with the reported metric and rewards models that win where interpolation is weak. That single change produced the n.s.→significant progression of Fig. 4; we document it as methodology, not as the scientific contribution.

8 Is it deployable?

Two things stand between a skill score and a usable instrument: it has to run inside the measurement cadence, and it has to say how much to trust it. Neither was previously reported, and both turn out to have answers that a reader would not guess.

Latency: it fits, on a CPU, and the GPU is the wrong device. CES sits on a 10 ms grid, so an online gap-filler has 10 ms per step minus whatever acquisition and control already spend. Measuring the network forward pass (feature assembly is outside the model and excluded), at batch 1 over 1,000 iterations after warmup, the tail—which is what decides whether a real-time loop holds its deadline—is **6.4 ms at p99 on CPU for W=4 (8.7 ms for W=2), with a median of 2.8 ms**: 64–87% of one grid period. On the same machine, **CUDA at batch 1 is $\approx 8\times$ slower** (median 21 ms, p99 43–72 ms), overrunning the budget several-fold. At 0.2M parameters there is nothing to amortize kernel-launch overhead against. We verified this two ways—per-call timing, which forces the device idle between calls and is the realistic model for a 10 ms loop, and amortized back-to-back timing—and they agree, so it is not an artifact of idling the GPU. The practical guidance is the opposite of the default: *run this model on the control computer's CPU.* Bulk reprocessing also favours CPU here (48k vs. 24k samples/s at batch 512), though that comparison is specific to a laptop GPU and a very small network and we do not generalize it.

Uncertainty: distribution-free intervals without touching the model. The community standard for fusion profile data is Gaussian-process regression, which returns a posterior; our model returns a point. Fitting a variance or quantile head would require retraining and would move the point predictions, confounding every number above—so we use *split conformal prediction* instead, which calibrates on the validation split, changes nothing about the predictor, and gives distribution-free finite-sample marginal coverage. Two variants: a single quantile (*global*), and one quantile per $\Delta t \times$ activity stratum (*Mondrian*), the same covariates as §6.4. The identical procedure is applied to PCHIP and persistence, so what is compared is interval quality, not calibration technique. At $\alpha = 0.10$:

- **The model's intervals beat both baselines' on every seed, target and variant** (8/8 each) by the Winkler interval score, which penalises width and misses jointly—roughly 0.80 of persistence's score. This does not follow from better point estimates; it is a separate property.
- **Stratifying helps**: Mondrian improves the interval score everywhere even though it *widens* the mean interval, i.e. it puts width where Δt and activity say it belongs.
- **The information asymmetry appears in the uncertainty too, weakly.** At matched coverage the model shrinks the interval to 0.884 of persistence's width for T_i but only 0.937 for V_{rot}—same direction as the point estimates, much smaller in magnitude. The model partially knows what it does not know.
- **Honest failure: coverage is marginal, not conditional.** T_i under-covers on two of four splits (87.0–88.9% against a 90% target), and per-shot coverage runs from ≈50–68% at the 10th percentile to 100% at the 90th. Conformal guarantees coverage under exchangeability, and calibration and test are disjoint *discharges*; shot-level distribution shift breaks it. For a deployed instrument this matters more than the pooled number, and fixing it needs shot-conditional calibration that the present shot count cannot support.

9 Where the remaining headroom is

The negative results in this paper are unusually specific, and specificity is what makes them useful: they do not say "this problem is hard", they say *which change would move it.* Capacity is ruled out (§4) and longer history is ruled out (§6.7). What remains is three concrete levers, each identified by evidence already in this dataset, and each a change to *measurement or coverage* rather than to the estimator.

1. Reach, not depth (largest immediate gain, T_i). Only 54.1% of genuinely missing T_i rows—and 4.8% of V_{rot} rows—have an observed value inside the W=4 window, so the method is inapplicable to the rest (§6.4). The window sweep shows the correct response is *not* a longer contiguous window, which buys no skill (§6.7), but a wider *reach*: going from W=2 to W=8 leaves total scored counts almost unchanged while growing the $\Delta t > 15\,\mathrm{ms}$ subset 456 → 1,958 and $\Delta t > 45\,\mathrm{ms}$ 14 → 135. Decoupling the two—keeping two or three history *slots* but drawing them from a wider span—should extend the domain without paying the capacity or distribution-shift cost of a long window. Since skill is flat in history length, this is close to a free extension of applicability, and it is the one change that directly enlarges the fraction of the real problem addressed.

2. The Mirnov information was destroyed by preprocessing, not absent from the plasma. It would be wrong to conclude from §6.6 that magnetics carry no rotation information. On the identical 10 ms grid, lag-1 autocorrelation within contiguous blocks is +0.568 for BES and +0.572 for ECEI but −0.009 for the Mirnov coils, with 82% of blocks below $|r| = 0.1$: the magnetic channel is white noise *on this grid.* That is the signature of $\mathrm{d}B/\mathrm{d}t$ oscillating at the kHz mode frequency and being decimated to 100 Hz without an anti-aliasing filter, so successive samples have random relative phase. The mode-rotation frequency—the one quantity in our diagnostic set that could proxy toroidal rotation—was therefore discarded upstream of the model, which is also why every derived MC feature we tried (integral, PCHIP integral, $|MC|$, rolling RMS) failed: information already lost cannot be recovered downstream. The fix is a preprocessing change, not a model change: per-window RMS, band power and mode number computed from the *raw* kHz Mirnov timeseries. This is the highest-value experiment we can name for V_{rot}, and it is testable on archived data.

3. The actuator channel is simply absent. Toroidal rotation is set by injected torque, and no torque signal exists in this dataset. The data show the break precisely: between shots, the ECE-derived T_e surrogate correlates with T_i at $r = +0.353$ ($p = 3 \times 10^{-17}$) but with V_{rot} at $r = +0.024$ ($p = 0.58$), and within shots the T_i correlation keeps a consistent sign while the V_{rot} one is sign-random. So T_e does track the heating level, and heating level does not reach rotation—power is not torque, which depends on beam energy, tangency radius and injection geometry. Adding NBI torque (or the per-beam power and geometry from which it is computed) is a data-acquisition task, not a modelling one, and it is the only lever we can identify that addresses the *cause* of rotation rather than its autocorrelation. A positive control already exists in the literature: recurrent full-discharge simulators on DIII-D that receive the beam actuators as inputs do predict rotation's evolution [51], so the missing channel is learnable once measured.

None of these is a statement that the present result is at a ceiling. They are the places where the next measurable gain is, ordered by how cheaply they can be tried, and all three are actionable on archived KSTAR data.

10 Limitations

Statistical power. The unit of replication is the shot; with ≈96 (T_i) / 91 (V_{rot}) test shots, power bounds every significance call, and per-shot squared-error differences are heavy-tailed (a few discharges dominate the bootstrap spread). **MNAR optimism.** Skill is measured on observed points only; genuinely missing points may be harder. **The offline claim is bounded above by a GP tie.** A post-hoc GP arm (§6.2) beats the pre-registered PCHIP on every split and the model only ties it, so "beats future-using interpolation" must be read as "beats the pre-registered interpolants"; it does not extend to every offline method. **CES fit-failure artifacts deflate rather than inflate the headline.** Spectral-fit failures ($T_i > 3\,\mathrm{keV}$; global p99 $= 2{,}089\,\mathrm{eV}$) remain in the pre-registered population. A paired sensitivity check that drops them (0.4–0.6% of rows, removed from all arms alike) keeps 4/4 PASS and roughly *doubles* the skill (+0.18–+0.28 → +0.36–+0.59): the spikes are unpredictable for every arm and drag the MSE ratio toward one, so the headline is conservative with respect to this artifact. **Metric asymmetry.** Interpolants use full-shot neighbors while the model sees a $W{=}4$ history; this is deliberately adverse to the model but complicates direct interpretation. **Single device and model family.** Window sensitivity is now characterized (§6.7) and architecture sensitivity partially so—a full-grid sequence reframing under the paired protocol—but all of it is on one device and one late-fusion family. Nothing here establishes transfer to another tokamak's diagnostic set. We also report against our own headline model: a full-grid sequence variant that combines the reframing with the same observation-blocked V_{rot} routing, trained hold-free with per-shot standardization, reaches a *higher* T_i skill than the pre-registered model on all four splits (paired mean +0.045, significant on 1/4) at roughly a tenth of the training cost. We do not restate the headline around it—the pre-registered protocol names the model in advance, and one-of-four significance does not carry a claim—but it is the direction we would build on next. **Campaign transfer was a real limitation, and its named repair now works** (§6.5): trained on earlier discharges the model did not beat offline interpolation on later ones (0/4), and the identified cause—fast-diagnostic drift against train-file-only normalization—is repaired by per-shot standardization (+0.155 mean paired T_i gain, 4/4 initialisations, no significant headline cost). What remains untested is the deployable form of that repair: an online estimator cannot see a discharge's future, so a causal running (expanding-window/EWMA) variant is the version that would actually ship. **Uncertainty is calibrated marginally, not conditionally** (§8): per-shot coverage ranges from ≈50% to 100%, so a per-discharge trust statement is not yet supported. **Large-gap bins** are now pooled across the four splits and carry shot-clustered CIs (§6.3), but the widest bin still holds 167 samples over 62 shots, and the model is significantly worse than future-using PCHIP there. **Scope.** The pedestal-top framing is inherited from the data selection (pedestal-top CES channels); no radial-dependence experiment is performed, and event-phase analyses (ELM cycle, L–H transitions) are future work—the high-variability analysis of §6.9 is an event-agnostic proxy.

11 Conclusion

We built a causal, leakage-hardened multimodal nowcaster for sparse KSTAR CES targets and evaluated it under a pre-registered, shot-clustered protocol against a deliberately adverse bar: interpolation that sees the future. On the observed population the model beats future-using PCHIP for ion temperature with significant, replicated skill (+0.18–+0.28 across four independent splits) — while tying the strongest offline smoother, a post-hoc GP arm (§6.2) — and it beats every causal baseline for both targets by a wide margin, including in the non-adjacent regime ($\Delta t > 15\,\text{ms}$) where earlier readings had lacked the power to say anything.

The claim we are willing to make about deployment is narrower and better supported. Two independent stress tests—reweighting onto the covariate distribution of the genuinely missing points, and re-splitting strictly in time across campaigns—agree: the advantage over *causal* methods survives both (+0.29 and +0.22), while the advantage over *offline* interpolation survives neither (1/4 and 0/4). Those are different claims about different opponents, and conflating them is the main way a result like this gets oversold. An online virtual sensor competes with persistence, not with an interpolant that reads the future; by that comparison the nowcaster works at the points where it matters, with the MNAR correction costing only 0.06–0.12 of skill. It also runs inside the measurement cadence (p99 6.4 ms on CPU) and emits calibrated intervals that beat both baselines' by a proper scoring rule.

We also report where it does not work and why. At W=4 only 54% of missing T_i rows and 4.8% of missing V_{rot} rows are even in the model's domain; beyond 105 ms two-sided interpolation genuinely wins; rotation ties interpolation globally; the predictive intervals are calibrated marginally but not per discharge; and across a campaign boundary the offline-interpolation advantage disappears entirely. None of these is a ceiling argument, and none is left as a shrug. Each points at a specific, testable change: wider history *reach* at constant depth; per-window features recomputed from the raw kHz Mirnov stream whose information 100 Hz decimation destroyed before the model ever saw it; the NBI torque channel that is absent from the dataset but is the physical cause of rotation (§9). The fourth, per-shot standardization of the fast diagnostics, we no longer list as a proposal: it was the repair the campaign analysis named, and running it recovers +0.155 mean paired T_i skill across campaigns on 4/4 initialisations with no significant cost on the headline split (§6.5) — a named negative result converted into a measured gain. Alongside these, the work contributes a corrected V_{rot} evaluation (54% instrument holds, which we show also contaminate training), observation-masked attention pooling, a complexity ladder that prices the model's opacity at 31.5% of the T_i margin recoverable by 1,258 interpretable parameters, and an architecture-selection protocol gated on clean interpolation-relative skill.

Reproducibility

Splits are pinned to disk and verified on reload; normalization statistics are train-file-only; model selection never reads test data; bootstrap uses a fixed seed and 10,000 resamples; all headline numbers are read from frozen evaluation artifacts.

Code availability

The full pipeline, the controlled-experiment runners behind every result reported here, the pinned train/validation/test splits, and the collector that regenerates every quoted number from the frozen evaluation artifacts are openly available at `https://github.com/iseungsang01/ml-intern-revision`. The published architecture is committed as `ces_prediction/model_iter009.py` and guarded by a content hash in the test suite, so the code that produced these numbers cannot drift from the code that is released.

Data availability

The KSTAR diagnostic data underlying this study — 641 discharges, shots 30801–32751 — are not redistributed with the code. They are experimental data of the KSTAR device and are governed by the operating institution's data policy; access requests should be directed there. The repository contains everything needed to reproduce the analysis once the shot files are in place: the exact file-level split manifests, the environment configuration for every reported run, and the per-sample squared-error archives from which the bootstrap statistics are computed.

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